

Simple Linear Regression

A Crash Course from First Principles

Linear Statistical Models — Lectures 1–4 (20 Jul – 29 Jul)

This is the first of four crash courses. It assumes you know only basic linear algebra: vectors, matrices, transpose, inverse, and what a subspace is. Everything statistical is built from scratch. The companions are *Matrix Algebra and the Gauss–Markov Framework*, *Quadratic Forms, Nested Models and Testing*, and *One-Way ANOVA*.

Contents

1	Why this exists	1
1.1	The three assumptions, and what each one buys	2
1.2	Why squares? (three fitting criteria)	2
2	The least squares estimates	3
2.1	$\hat{\beta}_1$ is a weighted sum of the Y_i — the key observation	3
2.2	Fitted values, residuals, and the estimate of σ^2	3
2.3	Least squares versus maximum likelihood	4
3	The ANOVA decomposition, and why it is Pythagoras	4
3.1	The geometry — learn this picture, it runs the whole course	4
4	The hat matrix	5
4.1	Unbiasedness and variance, done in matrix form	5
5	Distribution theory (now we need normality)	6
6	Tests for the regression coefficients	7
6.1	The idea of a pivot	7
6.2	Test for the slope (test for regression effect)	7
6.3	Test for a general linear combination	7
6.4	Joint test for both parameters at once	8
6.5	Bonferroni: simultaneous testing done crudely but safely	8
7	Confidence intervals, and the duality with testing	8
7.1	The intervals you need	8
7.2	Confidence <i>sets</i> for the pair β : ellipse versus rectangle	9
8	Prediction: the interval that is wider than you expect	9
9	Influence of a single observation (leave-one-out)	10
10	Practice, with answers	10

1 Why this exists

You have pairs $(x_1, Y_1), \dots, (x_n, Y_n)$. The x_i are known numbers you chose or observed; the Y_i are random. You want to say something about how Y depends on x .

The honest object of interest is the whole conditional distribution of Y given $X = x$. That is far too much to estimate from n points. So we make a decision that governs this entire course:

The one modelling decision

We model only the *conditional mean*, and we assume it is a straight line:

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x.$$

Everything else about the distribution is dumped into an error term.

A full regression model actually specifies three things:

- (a) the conditional *mean* of Y given X — this is the line;
- (b) the conditional *variability* — we will take it constant, σ^2 ;
- (c) the conditional *distribution* — we will take it normal, but only when we need it.

Writing $\varepsilon_i := Y_i - (\beta_0 + \beta_1 x_i)$ turns this into

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n. \quad (1)$$

1.1 The three assumptions, and what each one buys

Assumption	What it is for
(1) $\mathbb{E}(\varepsilon_i \mid x_i) = 0$	Identifiability. Without it, β_0 and a constant shift in ε are indistinguishable. Makes the LS estimates unbiased.
(2) $\text{Var}(\varepsilon_i \mid x_i) = \sigma^2$	Homoscedasticity. Gives the clean variance formulas and makes $\hat{\sigma}^2$ unbiased.
(3) $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0, i \neq j$	Uncorrelatedness. Also needed for the variance formulas.
★ $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$	Normality. Only needed for t -tests, F -tests and confidence intervals. <i>Not</i> needed for unbiasedness or for Gauss–Markov.

Common confusion: “do I need normality here?”

This is worth being clear about because it is a cheap mark in the exam. Unbiasedness of $\hat{\beta}$ and $\hat{\sigma}^2$, the variance formulas, and the whole Gauss–Markov theorem need only (1)–(3). Normality enters *only* when you want a distribution — the moment you write t_{n-2} or χ_{n-2}^2 or F , you have used ★.

1.2 Why squares? (three fitting criteria)

The professor listed three ways to fit a line:

$$\sum (Y_i - \beta_0 - \beta_1 x_i)^2, \quad \sum |Y_i - \beta_0 - \beta_1 x_i|, \quad \sum |Y_i - \beta_0 - \beta_1 x_i|^\alpha, \quad \alpha \geq 1.$$

Least squares wins not because it is more “correct” but because it is the only one with a closed-form solution, a clean geometry (orthogonal projection), and an exact distribution theory. The other two require numerical optimisation.

Poisson regression: a model where the three parts are linked

The notes give the contrast: $Y \mid X = x \sim \text{Poisson}(\beta_0 + \beta_1 x)$. Here

$$\mathbb{E}(Y \mid X = x) = \text{Var}(Y \mid X = x) = \beta_0 + \beta_1 x.$$

The mean is still linear, but assumption (2) fails — the variance changes with x . This is the standard example that the three ingredients (mean / variance / distribution) really are separate choices.

2 The least squares estimates

Minimise $Q(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$. Set the two partial derivatives to zero:

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum (Y_i - \beta_0 - \beta_1 x_i) = 0, \quad \frac{\partial Q}{\partial \beta_1} = -2 \sum x_i (Y_i - \beta_0 - \beta_1 x_i) = 0.$$

These are the **normal equations**. The first gives $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ immediately. Substituting into the second and simplifying:

The formulas

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_i (x_i - \bar{x})^2} = \frac{S_{xY}}{S_x^2} = r_{xY} \cdot \frac{s_Y}{s_x}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

2.1 $\hat{\beta}_1$ is a weighted sum of the Y_i — the key observation

Because $\sum (x_i - \bar{x})\bar{Y} = \bar{Y} \sum (x_i - \bar{x}) = 0$, the $\bar{Y}$ in the numerator does nothing, so

$$\hat{\beta}_1 = \sum_{i=1}^n c_i Y_i, \quad c_i = \frac{x_i - \bar{x}}{\sum_j (x_j - \bar{x})^2}. \quad (2)$$

This is the single most useful fact in the whole topic. It says $\hat{\beta}_1$ is a **linear** function of the data with *non-random* coefficients. Consequences, all immediate:

- $\mathbb{E}\hat{\beta}_1 = \sum c_i \mathbb{E}Y_i = \sum c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum c_i + \beta_1 \sum c_i x_i = 0 + \beta_1$. (Check: $\sum c_i = 0$ and $\sum c_i x_i = \sum c_i (x_i - \bar{x}) = 1$.) So it is unbiased.
- $\text{Var} \hat{\beta}_1 = \sigma^2 \sum c_i^2 = \sigma^2 / \sum (x_i - \bar{x})^2 = \sigma^2 / S_x^2$, using uncorrelatedness.
- If the Y_i are normal, then $\hat{\beta}_1$ is normal, being a linear combination of independent normals.

The variance formulas you must be able to write down cold

With $S_x^2 := \sum_i (x_i - \bar{x})^2$ (note: *not* divided by n):

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_x^2}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_x^2} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_x^2}.$$

All three come out of one matrix computation, done in §4.

2.2 Fitted values, residuals, and the estimate of σ^2

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \quad (\text{estimates } \mathbb{E}(Y \mid X = x_i)), \quad e_i = Y_i - \hat{Y}_i \quad (\text{estimates } \varepsilon_i).$$

The two normal equations say exactly

$$\sum_i e_i = 0 \quad \text{and} \quad \sum_i e_i x_i = 0.$$

That is *two* linear constraints on the n residuals, so the residual vector lives in an $(n-2)$ -dimensional space. Hence

Why $n - 2$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n - 2} = \frac{\text{SSE}}{n - 2} =: \text{MSE},$$

and under (1)–(3), $\mathbb{E}\hat{\sigma}^2 = \sigma^2$. The $n - 2$ is the *degrees of freedom of the residual vector* — n coordinates minus 2 estimated parameters. In the general model with p parameters it becomes $n - p$; in one-way ANOVA with r groups it becomes $n - r$.

2.3 Least squares versus maximum likelihood

Add normality: $Y_i \mid x_i \stackrel{\text{indep}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$. The log-likelihood is

$$\ell(\beta_0, \beta_1, \sigma^2) = c_n - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (Y_i - \beta_0 - \beta_1 x_i)^2.$$

For any fixed σ^2 , maximising over (β_0, β_1) means *minimising the sum of squares*. So

LS = MLE for the slope and intercept, but not for σ^2

$$(\hat{\beta}_0^{ML}, \hat{\beta}_1^{ML}) = (\hat{\beta}_0^{LS}, \hat{\beta}_1^{LS}), \quad \hat{\sigma}_{ML}^2 = \frac{1}{n} \sum e_i^2 = \frac{n - 2}{n} \hat{\sigma}^2.$$

The MLE of σ^2 divides by n , so it is **biased downwards**. This is a one-line exam question.

3 The ANOVA decomposition, and why it is Pythagoras

Define

$$\text{SST} = \sum (Y_i - \bar{Y})^2, \quad \text{SSR} = \sum (\hat{Y}_i - \bar{Y})^2, \quad \text{SSE} = \sum (Y_i - \hat{Y}_i)^2.$$

(SST = total variability; SSR = variability explained by the regression; SSE = leftover.)

ANOVA identity

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}$$

Proof. (the algebraic version from the notes) Start from $\hat{\beta}_1^2 S_x^2$:

$$\hat{\beta}_1^2 \sum_i (x_i - \bar{x})^2 = \sum_i \{\hat{\beta}_1(x_i - \bar{x})\}^2 = \sum_i \{(\hat{\beta}_0 + \hat{\beta}_1 x_i) - (\hat{\beta}_0 + \hat{\beta}_1 \bar{x})\}^2 = \sum_i (\hat{Y}_i - \bar{\hat{Y}})^2 = \text{SSR},$$

using $\bar{\hat{Y}} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{Y}$. Then the direct expansion $\sum e_i^2 = \sum (Y_i - \bar{Y})^2 - \hat{\beta}_1^2 \sum (x_i - \bar{x})^2$ (which you get by writing $e_i = (Y_i - \bar{Y}) - \hat{\beta}_1(x_i - \bar{x})$ and squaring, the cross term being $-2\hat{\beta}_1 S_{xY} = -2\hat{\beta}_1^2 S_x^2$) gives $\text{SSE} = \text{SST} - \text{SSR}$. ■

3.1 The geometry — learn this picture, it runs the whole course

Think of $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ as one point in $\mathbb{R}^n$. The model says $\mathbb{E}\mathbf{Y} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, i.e. $\mathbb{E}\mathbf{Y}$ lies in the two-dimensional subspace

$$\mathcal{C}([\mathbf{1} : \mathbf{x}]) = \{a\mathbf{1} + b\mathbf{x} : a, b \in \mathbb{R}\} \subset \mathbb{R}^n.$$

Least squares = orthogonal projection

Minimising $\|\mathbf{Y} - a\mathbf{1} - b\mathbf{x}\|^2$ over (a, b) is precisely asking for the *closest point* in that plane to $\mathbf{Y}$. The answer is the orthogonal projection $\hat{\mathbf{Y}}$, and the residual $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ is perpendicular to the plane:

$$\mathbf{e}^T \mathbf{1} = 0, \quad \mathbf{e}^T \mathbf{x} = 0.$$

These *are* the normal equations. Least squares is not a formula; it is a right angle.

Pythagoras in the right triangle with legs $\hat{\mathbf{Y}}$ and $\mathbf{e}$:

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.$$

Subtract $n\bar{Y}^2$ from both sides (legitimate, since $\overline{\hat{Y}} = \bar{Y}$):

$$\underbrace{\|\mathbf{Y}\|^2 - n\bar{Y}^2}_{\text{SST}} = \underbrace{\|\hat{\mathbf{Y}}\|^2 - n\bar{Y}^2}_{\text{SSR}} + \underbrace{\|\mathbf{e}\|^2}_{\text{SSE}}.$$

4 The hat matrix

Stack the model: let $\mathbf{X}$ be the $n \times 2$ matrix $[\mathbf{1} : \mathbf{x}]$ and $\boldsymbol{\beta} = (\beta_0, \beta_1)^T$. Then (1) is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, and the normal equations $\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = 0$ give

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}, \quad \hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}} = H \mathbf{Y}, \quad H := \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.$$

H is the **hat matrix** (it puts the hat on $\mathbf{Y}$). It is the matrix of the orthogonal projection onto $\mathcal{C}(\mathbf{X})$.

Properties of H — all one-liners, all examinable

- $H^T = H$ (symmetric): clear from the formula, since $(\mathbf{X}^T \mathbf{X})^{-1}$ is symmetric.
- $H^2 = H$ (idempotent): $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$. Projecting twice is the same as projecting once.
- $H\mathbf{X} = \mathbf{X}$; in particular $H\mathbf{1} = \mathbf{1}$, since $\mathbf{1}$ is a column of $\mathbf{X}$. (Vectors already in the plane are left alone.)
- $\text{tr}(H) = \text{rank}(H) = 2$ (in general p). Because H is idempotent, its eigenvalues are 0 and 1, so trace = rank = dimension of the space projected onto.
- $I - H$ is the projection onto the orthogonal complement, so $(I - H)\mathbf{X} = 0$ and $\mathbf{e} = (I - H)\mathbf{Y} = (I - H)\boldsymbol{\varepsilon}$.
- With $P_1 := \frac{1}{n}\mathbf{1}\mathbf{1}^T$ (projection onto the constant vector), $HP_1 = P_1$, since P_1 projects into a subspace of $\mathcal{C}(\mathbf{X})$.

4.1 Unbiasedness and variance, done in matrix form

$\hat{\boldsymbol{\beta}}$ is unbiased:

$$\mathbb{E}\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{E}\mathbf{Y} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}.$$

Variance: writing $C = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$, and using $\text{Var}(\mathbf{Y}) = \sigma^2 I$,

$$\text{Var}(\hat{\boldsymbol{\beta}}) = C \text{Var}(\mathbf{Y}) C^T = \sigma^2 C C^T = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.$$

For simple linear regression,

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad (\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{n \sum (x_i - \bar{x})^2} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

because $n \sum x_i^2 - (\sum x_i)^2 = n \sum (x_i - \bar{x})^2 = n S_x^2$. Reading off the diagonal recovers exactly the boxed variance formulas of §2.

$\hat{\sigma}^2$ is **unbiased** — **the trace trick**. This proof is worth memorising because the same three lines appear again for ANOVA.

$$\begin{aligned} \mathbb{E}(\text{SSE}) &= \mathbb{E}\|(I - H)\mathbf{Y}\|^2 = \mathbb{E}\|(I - H)\boldsymbol{\varepsilon}\|^2 && \text{since } (I - H)\mathbf{X} = \mathbf{0} \\ &= \mathbb{E}[\boldsymbol{\varepsilon}^T (I - H) \boldsymbol{\varepsilon}] = \mathbb{E}[\text{tr}(\boldsymbol{\varepsilon}^T (I - H) \boldsymbol{\varepsilon})] && \text{a scalar equals its trace} \\ &= \mathbb{E}[\text{tr}((I - H) \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T)] = \text{tr}((I - H) \mathbb{E}[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T]) && \text{tr}(AB) = \text{tr}(BA) \\ &= \sigma^2 \text{tr}(I - H) = \sigma^2 \text{rank}(I - H) = \sigma^2(n - 2). \end{aligned}$$

Hence $\mathbb{E}[\text{SSE}/(n - 2)] = \sigma^2$. ■

Common confusion: $\sum (x_i - \bar{x})^2$ versus s_x^2

The notes write $S_x^2 = \sum (x_i - \bar{x})^2$ — the raw sum, no division. Some books write $s_x^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$. If you plug the wrong one into $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_x^2$ you get an answer off by a factor of $n - 1$. Whenever you see S_x^2 in this course, it is the *un-normalised* sum of squares.

5 Distribution theory (now we need normality)

Assume $\star$: $Y_i \stackrel{\text{indep}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$.

The three facts everything is built from

1. $\hat{\boldsymbol{\beta}} \sim \mathcal{N}_2(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1})$.
(Linear function of a normal vector.)
2. $\text{SSE} = (n - 2)\hat{\sigma}^2 \sim \sigma^2 \chi_{n-2}^2$.
3. $\hat{\boldsymbol{\beta}}$ and SSE are **independent**.

Why (2). $\text{SSE} = \boldsymbol{\varepsilon}^T (I - H) \boldsymbol{\varepsilon}$. Spectrally decompose the symmetric idempotent matrix, $I - H = P \Lambda P^T$ with $P P^T = P^T P = I$ and Λ diagonal with $n - 2$ ones and 2 zeros (eigenvalues of an idempotent matrix are 0/1; the number of ones is the rank). Put $U = P^T \boldsymbol{\varepsilon}$. Since $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$ and P is orthogonal, $U \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$ too. Then

$$\text{SSE} = U^T \Lambda U = \sum_{i=1}^{n-2} U_i^2 = \sigma^2 \sum_{i=1}^{n-2} \left(\frac{U_i}{\sigma} \right)^2 \sim \sigma^2 \chi_{n-2}^2.$$

Why (3). $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\varepsilon}$ and $\mathbf{e} = (I - H) \boldsymbol{\varepsilon}$ are jointly normal, so it suffices to show they are uncorrelated:

$$\mathbb{E}[(I - H) \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}] = \sigma^2 \underbrace{(I - H) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}}_{=0} = 0.$$

Since SSE is a function of $\mathbf{e}$ alone, it is independent of $\hat{\boldsymbol{\beta}}$. ■

The general lemma behind all of this

If $Z \sim \mathcal{N}_n(\mathbf{0}, I)$ and A is symmetric idempotent, then $Z^T A Z \sim \chi^2_{\text{rank}(A)}$. This is the engine; it is proved properly in Crash Course 3.

6 Tests for the regression coefficients

6.1 The idea of a pivot

To test $H_0 : \theta = \bar{\theta}$ we want a statistic whose distribution, *under* H_0 , does not depend on any unknown parameter. The natural candidate is

$$\frac{\hat{\theta} - \bar{\theta}}{s(\hat{\theta})},$$

where $s(\hat{\theta})$ is the estimated standard error — the true standard deviation with σ replaced by $\hat{\sigma}$. Such a quantity is called **pivotal**.

6.2 Test for the slope (test for regression effect)

$H_0 : \beta_1 = \bar{\beta}_1$ versus $H_1 : \beta_1 \neq \bar{\beta}_1$. Take

$$T_1 = \frac{\hat{\beta}_1 - \bar{\beta}_1}{s(\hat{\beta}_1)}, \quad s(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_x^2}} = \sqrt{\frac{\text{MSE}}{S_x^2}}.$$

Why it is t_{n-2} . Under H_0 ,

$$V_1 := \frac{\hat{\beta}_1 - \bar{\beta}_1}{\sigma / \sqrt{S_x^2}} \sim \mathcal{N}(0, 1), \quad \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2 \text{ independently,}$$

and

$$T_1 = \frac{V_1}{\sqrt{\hat{\sigma}^2 / \sigma^2}} = \frac{\mathcal{N}(0, 1)}{\sqrt{\chi_{n-2}^2 / (n-2)}} \sim t_{n-2},$$

which is the definition of a t distribution. **Reject H_0 if $|T_1| > t_{n-2}^{1-\alpha/2}$.**

The case $\bar{\beta}_1 = 0$ is the *test for lack of dependence on x* ; the case $\bar{\beta}_0 = 0$ (with the same construction using $s(\hat{\beta}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_x^2}}$) is the *test for no intercept*.

6.3 Test for a general linear combination

$H_0 : c_0\beta_0 + c_1\beta_1 = v$. Since $c_0\hat{\beta}_0 + c_1\hat{\beta}_1$ is again a linear function of a normal vector,

$$\text{Var}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1) = \sigma^2 (c_0 \ c_1)(\mathbf{X}^T \mathbf{X})^{-1} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix},$$

and

$$T = \frac{(c_0\hat{\beta}_0 + c_1\hat{\beta}_1) - v}{s(c_0\hat{\beta}_0 + c_1\hat{\beta}_1)} \stackrel{H_0}{\sim} t_{n-2}.$$

The important special case is $c_0 = 1$, $c_1 = x_h$: this tests a hypothesis about the value of the regression line at $x = x_h$.

6.4 Joint test for both parameters at once

$H_0 : \beta = \bar{\beta}$ (a statement about *both* coordinates). Under H_0 , $\hat{\beta} - \bar{\beta} \sim \mathcal{N}_2(\mathbf{0}, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1})$, so

$$\frac{(\hat{\beta} - \bar{\beta})^T (\mathbf{X}^T \mathbf{X}) (\hat{\beta} - \bar{\beta})}{\sigma^2} \sim \chi_2^2,$$

independently of $\hat{\sigma}^2 / \sigma^2 \sim \chi_{n-2}^2 / (n-2)$. Dividing,

$$W := (\hat{\beta} - \bar{\beta})^T (s^2(\hat{\beta}))^{-1} (\hat{\beta} - \bar{\beta}) = \frac{(\hat{\beta} - \bar{\beta})^T (\mathbf{X}^T \mathbf{X}) (\hat{\beta} - \bar{\beta})}{\hat{\sigma}^2} \stackrel{H_0}{\sim} 2 F_{2, n-2}.$$

Common confusion: the factor of 2 in $2F_{2, n-2}$

$F_{r, m}$ is defined as $\frac{\chi_r^2 / r}{\chi_m^2 / m}$ — both chi-squares divided by their degrees of freedom. The statistic W above has the numerator χ_2^2 *undivided*. So $W/2 \sim F_{2, n-2}$, i.e. $W \sim 2F_{2, n-2}$. In general with r constraints you get $W \sim r F_{r, n-p}$. Losing this factor is a classic slip.

6.5 Bonferroni: simultaneous testing done crudely but safely

Suppose you want a procedure that controls the error rate for *both* coefficients simultaneously. Let

$$A_0 = \left\{ \left| \frac{\hat{\beta}_0 - \bar{\beta}_0}{s(\hat{\beta}_0)} \right| > t_{n-2}^{1-\alpha_0/2} \right\}, \quad A_1 = \left\{ \left| \frac{\hat{\beta}_1 - \bar{\beta}_1}{s(\hat{\beta}_1)} \right| > t_{n-2}^{1-\alpha_1/2} \right\},$$

so $\mathbb{P}_{H_0}(A_j) = \alpha_j$. Then by the union bound (no independence needed!)

$$\mathbb{P}_{H_0}(A_0 \cup A_1) \leq \alpha_0 + \alpha_1.$$

Choosing $\alpha_0 = \alpha_1 = \alpha/2$ gives overall level $\leq \alpha$. That is the **Bonferroni method**. It is conservative — the true level is usually below α — but it needs no assumption about the dependence between the two tests.

7 Confidence intervals, and the duality with testing

Duality — one idea, two languages

Let $A_\alpha(\bar{\theta})$ be a level- α *acceptance region* for testing $H_0 : \theta = \bar{\theta}$, i.e. $\mathbb{P}_{\bar{\theta}}(\mathbf{Y} \in A_\alpha(\bar{\theta})) \geq 1 - \alpha$ for every $\bar{\theta}$. Define

$$C_\alpha(\mathbf{Y}) := \{\theta : \mathbf{Y} \in A_\alpha(\theta)\}.$$

Then $\mathbb{P}_\theta(\theta \in C_\alpha(\mathbf{Y})) \geq 1 - \alpha$ for every θ — so $C_\alpha(\mathbf{Y})$ is a $(1 - \alpha)$ -level confidence set. And conversely.

In words: a confidence set is exactly the set of null values you would not reject.

The prototype. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, testing $H_0 : \theta = \bar{\theta}$:

$$A_\alpha(\bar{\theta}) = \{\mathbf{x} : |\sqrt{n}(\bar{X} - \bar{\theta})| \leq z_{1-\alpha/2}\} \longleftrightarrow C_\alpha(\mathbf{X}) = \{\theta : |\sqrt{n}(\bar{X} - \theta)| \leq z_{1-\alpha/2}\}.$$

Same inequality, read as a statement about $\mathbf{x}$ or about θ .

7.1 The intervals you need

Inverting each t -test of §6 gives, at level $1 - \alpha$ with $t^* := t_{n-2}^{1-\alpha/2}$:

$$\beta_0 : \hat{\beta}_0 \pm t^* s(\hat{\beta}_0), \quad \beta_1 : \hat{\beta}_1 \pm t^* s(\hat{\beta}_1), \quad \beta_0 + \beta_1 x_h : \hat{\beta}_0 + \hat{\beta}_1 x_h \pm t^* s(\hat{\beta}_0 + \hat{\beta}_1 x_h).$$

7.2 Confidence sets for the pair β : ellipse versus rectangle

Approach I (exact, F -based). Invert the joint W -test:

$$C_\alpha(\mathbf{Y}) = \left\{ \beta \in \mathbb{R}^2 : (\beta - \hat{\beta})^T (\mathbf{X}^T \mathbf{X}) (\beta - \hat{\beta}) \leq 2\hat{\sigma}^2 F_{2,n-2}^{1-\alpha} \right\}.$$

This is an **ellipse** centred at $\hat{\beta}$. It is exact and has the smallest area, but it is tilted (because $\hat{\beta}_0$ and $\hat{\beta}_1$ are correlated).

Approach II (Bonferroni). Invert the two separate t -tests, each at level $\alpha/2$:

$$C_\alpha(\mathbf{Y}) = [\hat{\beta}_0 \pm t_{n-2}^{1-\alpha/4} s(\hat{\beta}_0)] \times [\hat{\beta}_1 \pm t_{n-2}^{1-\alpha/4} s(\hat{\beta}_1)].$$

This is a **rectangle**. It is conservative (coverage $\geq 1 - \alpha$), but it is easy to compute and easy to report coordinate by coordinate.

Common confusion: $\alpha/4$, not $\alpha/2$

Each of the two tests gets level $\alpha/2$; a two-sided test at level $\alpha/2$ uses the $1 - \alpha/4$ quantile. So the correct quantile in the Bonferroni rectangle is $t_{n-2}^{1-\alpha/4}$.

8 Prediction: the interval that is wider than you expect

Two genuinely different questions at a new point x_h :

1. **Estimation.** What is $\mathbb{E}(Y \mid X = x_h) = \beta_0 + \beta_1 x_h$? — a fixed number.
2. **Prediction.** What will $Y_{\text{new}} = \beta_0 + \beta_1 x_h + \varepsilon_{\text{new}}$ actually be? — a random variable not yet observed.

The point estimate is the same, $\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$, but the uncertainty is not.

Why $\hat{Y}_h$ answers both

$\mathbb{E}(Y \mid X)$ is characterised as the minimiser of $\mathbb{E}[(Y - g(X))^2]$ over all functions g with $\mathbb{E}[g(X)^2] < \infty$. So the best mean-squared predictor of a new Y is the conditional mean — the same target. What differs is the error you make.

Variance of the fitted value. Rewrite $\hat{Y}_h$ in the decoupled form

$$\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h = (\hat{\beta}_0 + \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 (x_h - \bar{x}) = \bar{Y} + \hat{\beta}_1 (x_h - \bar{x}).$$

Now $\bar{Y}$ and $\hat{\beta}_1$ are *uncorrelated*: $\text{Cov}(\bar{Y}, \hat{\beta}_1) = \sigma^2 \sum c_i / n = 0$ by (2). Hence there is no cross term and

$$\text{Var}(\hat{Y}_h) = \frac{\sigma^2}{n} + \frac{\sigma^2 (x_h - \bar{x})^2}{S_x^2} = \sigma^2 \left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right).$$

This is the tidy derivation the notes mark with “# How”. Note it is smallest at $x_h = \bar{x}$ and grows quadratically as you move away — the fitted line pivots about $(\bar{x}, \bar{Y})$.

Prediction error. $Y_{\text{new}} - \hat{Y}_h = \varepsilon_{\text{new}} + \{(\beta_0 + \beta_1 x_h) - (\hat{\beta}_0 + \hat{\beta}_1 x_h)\}$, and the two pieces are *independent* (ε_{new} is a new observation, not used in fitting). So the variances add:

$$\text{Var}(Y_{\text{new}} - \hat{Y}_h) = \sigma^2 + \text{Var}(\hat{Y}_h) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right).$$

Confidence interval versus prediction interval — the one-symbol difference

$$\text{CI for } \mathbb{E}(Y | x_h) : \quad \hat{Y}_h \pm t_{n-2}^{1-\alpha/2} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$$

$$\text{PI for } Y_{\text{new}} : \quad \hat{Y}_h \pm t_{n-2}^{1-\alpha/2} \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$$

The prediction interval has an extra “1+” and is therefore always wider. As $n \rightarrow \infty$ the CI shrinks to zero width but the PI does *not* — it converges to $\pm z_{1-\alpha/2}\sigma$, because you can never predict a fresh coin toss exactly.

Validity: $Y_{\text{new}} - \hat{Y}_h$ is normal with mean 0, and is independent of SSE (because ε_{new} is new and $\hat{\beta} \perp \text{SSE}$), so the standardised prediction error is again t_{n-2} .

9 Influence of a single observation (leave-one-out)

Let $(\hat{\beta}_0, \hat{\beta}_1)$ use all n points and $(\hat{\beta}_0^{(-n)}, \hat{\beta}_1^{(-n)})$ use the first $n-1$. The notes record the update formula

$$\hat{\beta}_1 = \hat{\beta}_1^{(-n)} + \frac{x_n - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} e_n^{(-n)}, \quad e_n^{(-n)} = Y_n - (\hat{\beta}_0^{(-n)} + \hat{\beta}_1^{(-n)} x_n).$$

Read it as: *the n -th point moves the slope by an amount proportional to (how surprising it was) $\times$ (how far its x is from the centre).* A point with $x_n = \bar{x}$ has no influence on the slope no matter how odd its Y ; a point far out in x has leverage. Also useful for the algebra:

$$\bar{Y} = \frac{n-1}{n} \bar{Y}^{(-n)} + \frac{1}{n} Y_n, \quad \bar{x} = \frac{n-1}{n} \bar{x}^{(-n)} + \frac{1}{n} x_n.$$

This is Homework 1, Question 3

HW1 Q3 is exactly this, in the “add an observation” direction, in matrix form:

$$\hat{\beta}^{(n+1)} - \hat{\beta}^{(n)} = c_{n+1} (Y_{n+1} - \hat{Y}_{n+1}^{(n)}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_{n+1}, \quad c_{n+1} = (1 + \mathbf{x}_{n+1}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_{n+1})^{-1}.$$

The tool is the Sherman–Morrison identity $(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}$ applied to $\mathbf{X}_{n+1}^T \mathbf{X}_{n+1} = \mathbf{X}^T \mathbf{X} + \mathbf{x}_{n+1} \mathbf{x}_{n+1}^T$. Worth a re-read tonight if you have time.

10 Practice, with answers

- P1.** Show that $\sum_i e_i = 0$ and $\sum_i e_i x_i = 0$, and deduce that $\widehat{\bar{Y}} = \bar{Y}$ and that $\mathbf{e}$ is orthogonal to $\hat{\mathbf{Y}}$.
- P2.** Prove $\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0$ under (1)–(3), and explain in one sentence why this makes the derivation of $\text{Var}(\hat{Y}_h)$ short.
- P3.** For the model with *no* intercept, $Y_i = \beta_1 x_i + \varepsilon_i$, derive the LS estimate of β_1 and its variance, and say what the residual degrees of freedom is.
- P4.** You are told $n = 20$, $\hat{\beta}_1 = 2.4$, $S_x^2 = 50$, $\text{SSE} = 72$. Test $H_0 : \beta_1 = 0$ at level 5% ($t_{18}^{0.975} = 2.101$).
- P5.** Explain, in terms of $\text{Var}(\hat{Y}_h)$, why extrapolating far outside the range of the observed x ’s is dangerous *even if the linear model is exactly true*.

Answers

P1. The two normal equations are $\sum(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$ and $\sum x_i(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$, which are literally $\sum e_i = 0$ and $\sum e_i x_i = 0$. Then $\hat{\bar{Y}} = \bar{Y} - \frac{1}{n} \sum e_i = \bar{Y}$. And $\mathbf{e}^T \hat{\mathbf{Y}} = \sum e_i(\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum e_i x_i = 0$.

P2. $\bar{Y} = \frac{1}{n} \sum Y_i$ and $\hat{\beta}_1 = \sum c_i Y_i$ with $c_i = (x_i - \bar{x})/S_x^2$. By uncorrelatedness, $\text{Cov}(\bar{Y}, \hat{\beta}_1) = \frac{\sigma^2}{n} \sum c_i = \frac{\sigma^2}{n S_x^2} \sum (x_i - \bar{x}) = 0$. Because $\hat{Y}_h = \bar{Y} + \hat{\beta}_1(x_h - \bar{x})$ is a sum of two *uncorrelated* terms, the variance is just the sum of the two variances — no cross term to compute.

P3. Minimising $\sum(Y_i - \beta_1 x_i)^2$ gives $\hat{\beta}_1 = \sum x_i Y_i / \sum x_i^2$, so $\text{Var}(\hat{\beta}_1) = \sigma^2 / \sum x_i^2$. Only one parameter is estimated, so there is one normal equation ($\sum e_i x_i = 0$, but *not* $\sum e_i = 0$), and the residual d.f. is $n - 1$; the unbiased estimate is $\hat{\sigma}^2 = \text{SSE}/(n - 1)$.

P4. $\hat{\sigma}^2 = 72/18 = 4$, so $s(\hat{\beta}_1) = \sqrt{4/50} = \sqrt{0.08} = 0.2828$. Then $T_1 = 2.4/0.2828 = 8.49$. Since $8.49 > 2.101$, reject H_0 decisively: there is a significant regression effect.

P5. $\text{Var}(\hat{Y}_h) = \sigma^2 \{ \frac{1}{n} + (x_h - \bar{x})^2 / S_x^2 \}$ grows *quadratically* in the distance of x_h from $\bar{x}$, and the prediction interval grows with it. Even with a perfectly correct model, the uncertainty in the estimated slope is levered up by the long moment arm: a small error in $\hat{\beta}_1$ translates into a large error far from the centre of the data. (And in practice the linear model is only trusted where you have data, which compounds the problem.)

Companions: Matrix Algebra and the Gauss–Markov Framework; Quadratic Forms, Nested Models and Testing; One-Way ANOVA.